

02_Referential Integrity Validation

Objective

The objective of this phase was to validate the structural integrity of the Olist dataset before performing data cleaning, feature engineering, and analytical operations.

Referential integrity validation ensured that primary keys, composite keys, and foreign key relationships were correctly maintained across all tables, preventing inaccurate joins, duplicate records, and misleading analytical results.

Primary Key Validation

Primary key candidates were evaluated to ensure uniqueness and suitability for identifying records within each entity.

Validated Primary Keys

Table	Primary Key
Customers	customer_id
Orders	order_id
Products	product_id
Sellers	seller_id
Category Translation	product_category_name

All validated primary keys were confirmed to contain unique values and were approved for relational analysis.

Composite Key Validation

Certain transactional tables required composite keys because a single column was insufficient to uniquely identify records.

Validated Composite Keys

Table	Composite Key
Order Items	order_id + order_item_id
Payments	order_id + payment_sequential

Both composite key structures were validated and confirmed to uniquely identify transaction-level records.

Foreign Key Relationship Validation

The dataset follows a relational marketplace structure where transactional tables are connected through shared identifiers.

The following relationships were validated:

Parent Table	Child Table	Relationship
Customers	Orders	customer_id
Orders	Payments	order_id
Orders	Reviews	order_id
Orders	Order Items	order_id
Products	Order Items	product_id
Sellers	Order Items	seller_id

Validation Results

Referential integrity checks were performed to identify orphan records and relationship mismatches across the dataset.

Results

- No orphan customer records found.
- No orphan order records found.
- No orphan payment records found.
- No orphan review records found.
- No orphan product records found.
- No orphan seller records found.

All core transactional relationships successfully passed validation.

Exception Identified

A small category translation mismatch was identified during validation.

Affected Categories

- pc_gamer
- portateis_cozinha_e_preparadores_de_alimentos

A total of 13 product records contained category values that were not available in the translation lookup table.

Business Impact

The issue was determined to have minimal analytical impact because:

- Only a small number of products were affected.
- Core transactional relationships remained intact.
- Revenue, customer, seller, and operational analyses were unaffected.

Outcome

The referential integrity validation confirmed that the Olist dataset maintains a reliable relational structure suitable for advanced analytics.

All critical relationships were successfully validated, enabling confident progression into data cleaning, feature engineering, exploratory analysis, SQL querying, and business intelligence development.